

Notes: Leary and Roberts (JF, 2005)

Do Firms Rebalance Their Capital Structures?

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1 Big picture

- **Question.** Do firms actively rebalance their capital structures toward an optimal leverage range, or are observed capital structures mainly the passive accumulation of market timing and stock-return shocks?
- **Core tension.** Prior evidence shows that leverage shocks are persistent. Baker and Wurgler interpret the persistence of past market valuations as evidence that capital structure reflects historical equity market timing. Welch argues that firms largely let stock returns determine leverage. Leary and Roberts argue that persistence alone does not reject rebalancing because adjustment is costly.
- **Main idea.** If issuing or retiring securities involves fixed, proportional, or convex adjustment costs, firms do not rebalance continuously. They tolerate deviations inside an optimal leverage region and adjust only when the benefits of adjustment exceed the costs.
- **Empirical approach.** The paper uses quarterly Compustat data and a duration model to study the timing of financing decisions. The model focuses on the hazard of a financing action - the probability that a firm adjusts now conditional on not having adjusted for a given number of quarters.
- **Main result.** Firms are inactive in most quarters, but when they do adjust, their choices are strongly consistent with active rebalancing. Firms are more likely to increase leverage when leverage is low or has fallen, and more likely to decrease leverage when leverage is high or has increased.
- **Important nuance.** Rebalancing occurs primarily through debt policy and, to a lesser extent, repurchases. Equity issuances appear more tied to valuation and investment-financing motives than to direct leverage rebalancing.
- **Economic takeaway.** Slow mean reversion and persistent leverage shocks can be the outcome of optimizing firms with adjustment costs. Persistence is not the same thing as indifference toward capital structure.

2 Incremental contribution of the paper

- **Reinterprets persistence evidence.** The paper shows that persistent effects of shocks on leverage are mechanically consistent with costly dynamic rebalancing. Therefore, the persistence documented by market timing and inertia papers is not decisive evidence against target leverage behavior.
- **Moves from leverage levels to actual financing actions.** Rather than infer corporate motives only from changes in leverage ratios, the paper directly studies discrete security issuances, retirements, repurchases, and leverage-increasing or leverage-decreasing financing events.
- **Introduces duration analysis to capital-structure dynamics.** The paper models the time between financing events. This is well suited to lumpy adjustment because financing decisions occur intermittently rather than every period.
- **Connects hazard shapes to adjustment cost structures.** The paper links fixed, proportional, and fixed-plus-convex adjustment costs to distinct hazard-curve shapes, then asks whether actual corporate financing hazards resemble these theoretical patterns.

- **Uses high-frequency quarterly data.** Quarterly data reveal that firms are inactive most of the time but adjust roughly once per year. Annual data can hide this spell structure and clustering.
- **Separates rebalancing from other motives.** The model includes standard tradeoff, pecking-order, agency, tax, investment, profitability, cash, valuation, and macro/industry controls. This lets the paper ask whether leverage itself predicts financing decisions after controlling for other determinants.
- **Challenges strong interpretations of market timing and inertia.** The paper does not deny that firms time equity issuances or that equity returns affect leverage. Instead, it argues that firms subsequently rebalance, often with delay because adjustment is costly.

3 Conceptual framework: adjustment costs and leverage dynamics

3.1 Why persistence does not reject rebalancing

A frictionless target-adjustment view predicts continuous rebalancing: after a shock, a firm immediately moves back to target leverage. If this were the right benchmark, persistent effects of equity issues or stock returns would be evidence against rebalancing. Leary and Roberts replace this frictionless benchmark with costly adjustment.

- **With adjustment costs, firms have a no-adjustment region.** Small leverage deviations are tolerated because the benefit of adjusting does not justify the issuance, retirement, or repurchase cost.
- **Shocks therefore persist.** A shock can move leverage but still leave it inside the no-adjustment region. No adjustment occurs until cumulative shocks make adjustment worthwhile.
- **Rebalancing is lumpy.** When a boundary is hit, firms adjust in discrete amounts. The size and frequency of adjustments depend on the cost function.

3.2 Three adjustment-cost regimes

1. **Fixed cost.** Once a firm pays the cost of adjustment, the marginal size of the adjustment is essentially free. The optimal adjustment is relatively large and infrequent. The firm tends to move back toward an interior point of the target region.
2. **Proportional cost.** Each dollar adjusted is costly. The firm makes small changes when it hits a boundary. This creates clusters of repeated adjustments because the firm remains close to the boundary after each small move.
3. **Fixed plus weakly convex cost.** The firm makes moderate adjustments. The fixed component makes adjustments lumpy; the convex component discourages very large adjustments. This generates adjustment patterns between the two extremes.

3.3 Implications for empirical work

- **Leverage levels are noisy indicators of policy.** Two firms following the same target-range policy can have different observed leverage ratios because they receive different shocks.

- **Partial adjustment coefficients can look slow.** If researchers estimate a continuous partial-adjustment model on data generated by lumpy adjustment, the estimated speed of adjustment may look very low even when firms actively rebalance.
- **The right object is the adjustment decision.** To understand motives, the paper studies why firms adjust when they adjust, not only how leverage evolves between adjustments.

4 Data and measurement

4.1 Sample

- **Data sources.** Quarterly Compustat data from 1984 to 2001, combined with CRSP monthly stock returns.
- **Exclusions.** Financial firms and utilities are removed because their financing policies are heavily affected by regulation.
- **Required variables.** Observations must have nonmissing book assets, equity issuances, equity repurchases, short-term debt, and long-term debt, because these variables are needed to classify financing events.
- **Dynamic panel restriction.** Firms must have at least four years of contiguous quarterly observations.
- **Final sample.** The main duration sample contains 127,308 firm-quarter observations for 3,494 firms, with firm time-series lengths ranging from 16 to 71 quarters.

4.2 Financing events

The paper defines “spikes” in financing activity using thresholds relative to lagged book assets.

- **Debt issue.** Change in total debt from quarter $t - 1$ to t , divided by lagged book assets, exceeds 5 percent.
- **Debt retirement.** Net reduction in debt, divided by lagged book assets, exceeds 5 percent.
- **Equity issue.** Net equity issuance exceeds 5 percent of lagged book assets.
- **Equity repurchase.** Net repurchase exceeds 1.25 percent of lagged book assets. A lower cutoff is used because repurchase programs are often smaller and spread over time.
- **Leverage increase.** Net debt issued minus net equity issued, divided by lagged book assets, exceeds 5 percent.
- **Leverage decrease.** Net equity issued minus net debt issued, divided by lagged book assets, exceeds 5 percent.

4.3 Why distinguish financing spikes from leverage adjustments?

A firm can issue both debt and equity in a way that leaves leverage nearly unchanged. Conversely, a financing action can have a strong directional effect on leverage. The leverage-increase and leverage-decrease variables are designed to isolate financing choices that move leverage, rather than simply identify gross external financing activity.

4.4 Key explanatory variables

- **Adjustment-cost proxies.** Estimated debt and equity underwriter spreads based on Altinkilic and Hansen; credit ratings; Altman Z-score; maximum turnover for equity repurchase constraints.
- **Rebalancing variables.** Current market leverage, change in leverage, whether a leverage decrease occurred earlier in the current leverage-increase spell (LeverDown), and whether a leverage increase occurred earlier in the current leverage-decrease spell (LeverUp).
- **Tradeoff controls.** Tangibility, volatility, selling expenses/product uniqueness, Z-score, depreciation and amortization, market-to-book ratio, profitability.
- **Pecking-order controls.** Cash balances, profitability, capital expenditures in quarter $t + 1$.
- **Timing and market controls.** Equity returns, year dummies, quarter dummies, and two-digit SIC dummies.

5 Empirical method: duration analysis

5.1 Hazard function

Let T be the time between financing events. The hazard function is

$$h(t) = \lim_{m \rightarrow 0} \frac{Pr(t < T < t + m \mid T > t)}{m}. \quad (1)$$

In the capital-structure setting, $h(t)$ is the instantaneous probability that a firm adjusts now, conditional on not having adjusted for t quarters.

- A high hazard means adjustment is likely soon.
- A low hazard means continued inactivity is likely.
- The shape of the hazard over time helps infer the structure of adjustment costs.

5.2 Semiparametric duration model

The estimated hazard for spell j of firm i is

$$h_{ij}(t \mid \omega_i) = \omega_i h_0(t) \exp\{x_{ij}(t)' \beta\}, \quad (2)$$

where:

- $h_0(t)$ is an unspecified baseline hazard, estimated as a step function;
- $x_{ij}(t)$ contains time-varying firm covariates;
- ω_i is firm-level unobserved heterogeneity, modeled with a gamma distribution;
- β describes how firm characteristics shift the baseline hazard proportionally.

5.3 Interpretation of coefficient estimates

The paper reports hazard impacts:

$$HI = 100 \times (\exp\{\hat{\beta}\} - 1). \quad (3)$$

A hazard impact is the percentage shift in the hazard curve caused by a one-unit increase in the covariate. For example, a hazard impact of 47 percent means that the covariate raises the conditional probability of adjustment by roughly 47 percent relative to the baseline.

6 Testable predictions

6.1 Predictions from adjustment costs

Cost structure	Predicted adjustment pattern	Predicted hazard shape
Fixed cost	Large, infrequent adjustments	Upward-sloping hazard
Proportional cost	Tiny, highly clustered adjustments	Steep downward-sloping hazard
Fixed plus convex cost	Moderate, clustered adjustments	Moderately downward-sloping hazard

6.2 Predictions from dynamic rebalancing

- If leverage is high, a firm should be more likely to decrease leverage and less likely to increase leverage.
- If leverage has recently increased, a firm should be more likely to decrease leverage.
- If leverage has recently decreased, a firm should be more likely to increase leverage.
- If the firm recently made a leverage-increasing action, future leverage-decreasing actions should become more likely than before.
- If the firm recently made a leverage-decreasing action, future leverage-increasing actions should become more likely than before.

6.3 Predictions from pecking-order considerations

- Firms with more cash and higher internal profitability should be less likely to use external finance.
- Firms with large investment needs should be more likely to use external finance.
- Equity issuance should be especially sensitive to valuation variables such as market-to-book and equity returns.

7 Figures and tables

7.1 Figure 1 and Figure 5: Adjustment costs and hazard shapes

Figure 1 read:

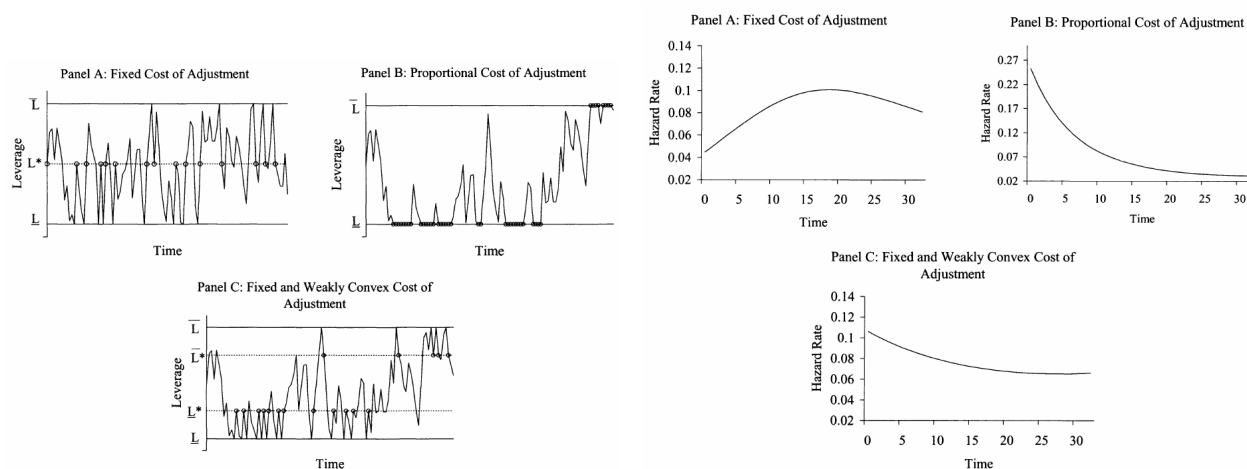
- Panel A shows a fixed-cost regime. Leverage wanders inside an optimal band and then jumps back toward the target when it hits a boundary. Adjustments are large and infrequent.

- Panel B shows proportional costs. When leverage hits a boundary, the firm adjusts only slightly, remaining close to the boundary and often adjusting again soon.
- Panel C shows fixed plus weakly convex costs. The firm makes intermediate-sized adjustments, moving back inside the band but not all the way to the initial target.

Figure 5 read:

- The simulated hazard under fixed costs rises with time since the last adjustment.
- The proportional-cost hazard starts high and falls sharply, reflecting clustered tiny adjustments right after a prior adjustment.
- The fixed-plus-convex hazard is downward-sloping but less steep, consistent with moderate clusters.

Economic message: The figures show why capital-structure dynamics can look persistent even when firms optimize. Costly adjustment creates no-adjustment regions and lumpy, clustered responses.



(a) Figure 1: Simulated leverage dynamics under different adjustment cost regimes.

Different cost functions imply different rebalancing sizes and frequencies.

(b) Figure 5: Simulated hazard curves under different adjustment cost regimes.

The hazard curve provides an empirical fingerprint of adjustment costs.

7.2 Figure 2 and Figure 3 Panel A: Rebalancing after equity issuance and EFWA

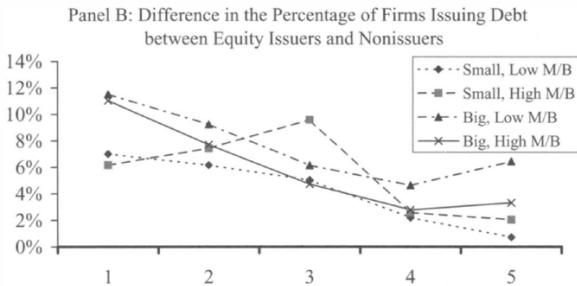
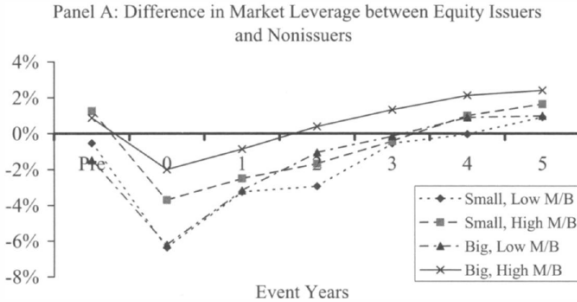
Figure 2 read:

- Panel A compares leverage of equity issuers and matched nonissuers by size and market-to-book portfolios.
- Equity issuers mechanically reduce leverage at issuance, but the leverage gap shrinks over the next few years.
- For large, low market-to-book firms, the issuer-nonissuer leverage gap falls from roughly 6.2 percent immediately after issuance to around 1 percent two years later.
- Panel B shows that equity issuers are more likely than nonissuers to issue debt later, indicating an active leverage-restoring response.

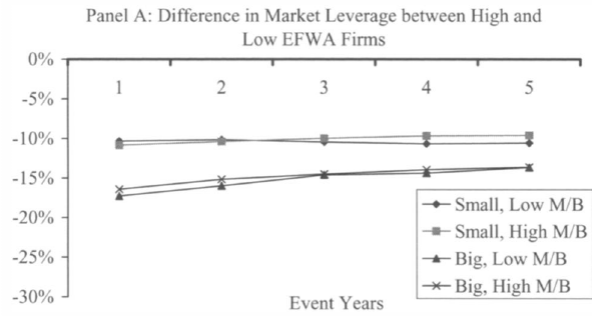
Figure 3 Panel A read:

- High-EFWA firms have persistently lower leverage than low-EFWA firms.
- This reproduces the intuition behind Baker and Wurgler's market-timing evidence.
- Leary and Roberts argue that this pattern is not necessarily evidence against rebalancing because EFWA is correlated with persistent market-to-book differences and adjustment costs.

Economic message: Firms that issue equity do not simply ignore leverage. They tend to issue debt later, but the response is gradual, consistent with costly rebalancing.



(a) Figure 2: Response to equity issuances. Equity issuers later raise leverage through debt issuance.



(b) Figure 3, Panel A: High versus low EFWA firms. High-EFWA firms have persistently lower leverage.

7.3 Figure 3 Panels B-D and Table I: Market timing versus adjustment costs

Figure 3 Panels B-D read:

- Panel B shows that firms with high historical average market-to-book ratios tend to have lower leverage.
- Panel C shows that the number of past equity issues by itself does not generate the same persistent leverage gap.
- Panel D shows that the size of past equity issues also explains less persistence than the broad historical market-to-book variable.
- These panels suggest that the Baker-Wurgler EFWA variable partly captures firms with persistently high market-to-book ratios rather than only cumulative equity timing events.

Table I read:

- Panel A replicates the Baker and Wurgler-style result: EFWA has a large negative coefficient in book leverage regressions.

- Panel B splits the sample by adjustment-cost proxies.
- The EFWA coefficient is smaller in magnitude for firms with lower estimated debt issuance costs. For estimated spreads, the coefficient moves from -10.04 in the high-cost group to -5.18 in the low-cost group.
- For credit ratings, firms with lower adjustment costs also show a smaller magnitude EFWA coefficient.
- This attenuation is exactly what dynamic rebalancing with adjustment costs predicts: firms with cheaper adjustment can undo historical leverage shocks faster.

Economic message: The market-timing variable is less persistent when adjustment is cheaper. This is hard to reconcile with pure indifference toward capital structure but natural under costly rebalancing.

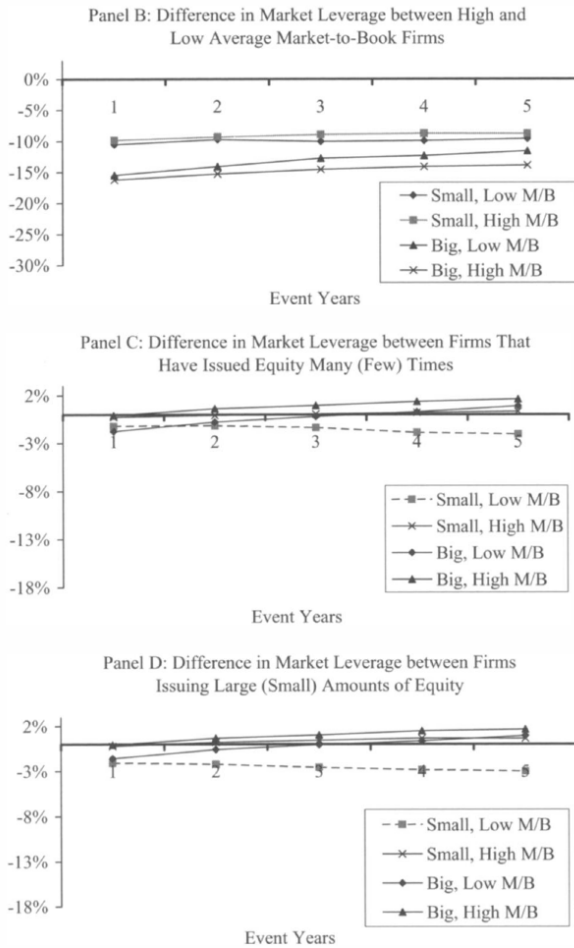


Table I

Persistence of Market-to-Book Effects

The sample is selected from annual COMPUSTAT data in a manner consistent with Baker and Wurgler (2002). Specifically, we start with all nonfinancial, nonutility firms listed on COMPUSTAT prior to 2000 and drop firms with missing values for book assets or with a minimum value for book assets of less than \$10 million. Panel A presents our replication of Baker and Wurgler's (2002) analysis ("All Firms" row of Panel A in Table III), which is a Fama-MacBeth regression in which a cross-sectional regression is run each year from 1980 through 1999 of book leverage in year t on the year $t-1$ values of the following variables: MA/B, defined as the ratio of total assets minus book equity plus market equity to total assets; EFWA, defined as the weighted average of MA/B from the first year COMPUSTAT reports market value data for that firm through year $t-1$, where the weights are the proportion of the firm's total external finance (net equity issued plus net debt issued) raised in each year; PPE/A, defined as net property, plant, and equipment divided by assets; EBITDA/A, defined as earnings before interest, taxes, and depreciation divided by assets; and, Size, defined as the log of net sales. Reported coefficients are the time-series average of the estimated cross-sectional coefficients. Panel B presents the results from running the same regression on subsamples determined by adjustment cost proxies: estimated underwriter spread, Altman's Z-score, and debt credit rating. Since the other coefficients are largely unaffected and our focus is on the impact of adjustment costs on the EFWA parameter, the other coefficients and t -statistics are suppressed. For the first two proxies, the portfolios are formed using the lower, middle, and upper third of the proxy distribution. For the credit rating proxy, we form the portfolio based on an above or below investment grade credit rating.

Panel A: Replication of Baker and Wurgler (2002) Results

	Intercept	EFWA	MA/B	PPE/A	EBITDA/A	Size						
	Coeff	t	Coeff	t	Coeff	t	r^2					
Full Sample	48.99	36.31	-8.07	-20.96	3.19	4.19	0.03	-0.58	-10.06	2.64	11.94	0.22

Panel B: Portfolios by Adjustment Costs

	EFWA		
	Coeff	t	
Cost Proxy	Low Cost	-5.18	-13.20
	Med Cost	-7.42	-12.97
	High Cost	-10.04	-16.18
Estimated spreads	Low Cost	-5.64	-10.15
	Med Cost	-8.94	-17.75
	High Cost	-8.15	-6.54
Z-score	Low Cost	-6.39	-16.34
	Med Cost	-9.32	-7.17
	High Cost	-9.35	-7.17
Credit rating	Low Cost	-5.18	-13.20
	Med Cost	-7.42	-12.97
	High Cost	-10.04	-16.18

(a) Figure 3, Panels B-D: Historical valuation and equity issue decompositions. (b) Table I: Persistence of market-to-book effects.

The persistent EFWA pattern is tied strongly to historical market-to-book. EFWA effects attenuate when adjustment costs are lower.

7.4 Figure 4 and Table II: Inertia, stock-return shocks, and simulated tradeoff behavior

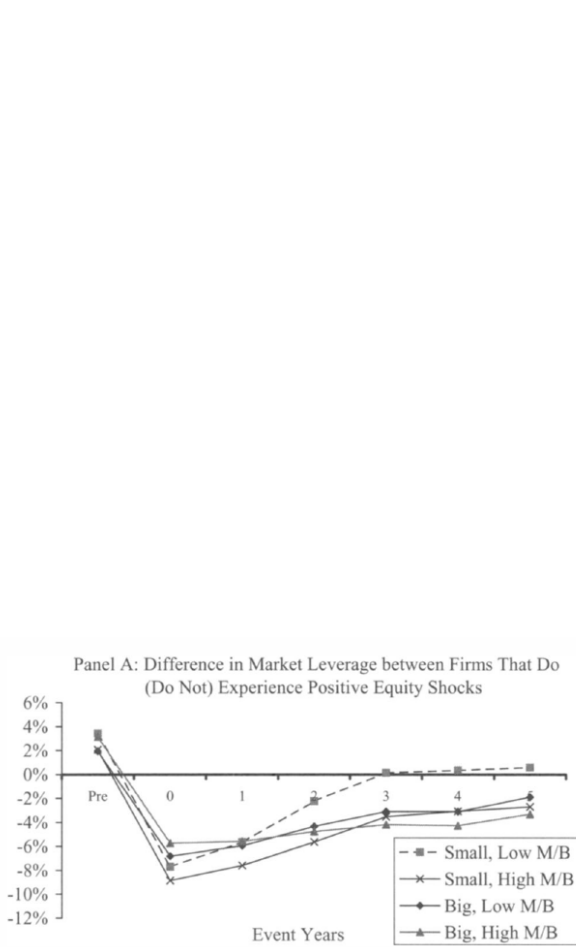
Figure 4 read:

- Panel A shows that positive equity shocks lower leverage relative to firms without such shocks.
- Panel B shows that firms experiencing positive shocks are more likely to issue debt later.
- Panel C shows that negative equity shocks raise leverage, but the effect fades over time.
- Panel D shows that firms experiencing negative shocks are more likely to retire debt later.
- Thus firms respond to equity price shocks, but not immediately.

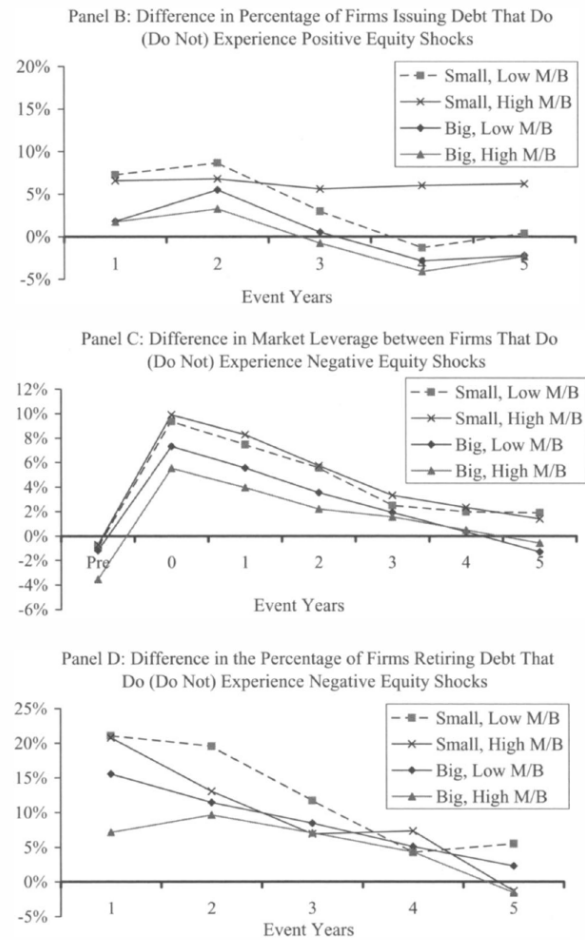
Table II read:

- The table reproduces Welch's leverage-on-stock-return regression results and then estimates the same model using simulated data from rebalancing models with adjustment costs.
- Welch's empirical results show α_2 near one at a one-year horizon and declining at longer horizons.
- Simulated data from models in which firms actively rebalance under adjustment costs produce very similar coefficient patterns.
- This means Welch's regression has limited power to distinguish inertia from costly dynamic tradeoff behavior.

Economic message: Stock returns affect leverage mechanically, but firms later use debt policy to offset the shocks. A regression that finds persistence can be replicated by a model in which firms do rebalance.



(a) Figure 4, Panel A: Positive equity shocks. Positive equity shocks reduce leverage initially.



(b) Figure 4, Panels B-D: Financing responses to equity shocks.

Debt issuance and retirement gradually offset equity shocks.

Table II
The Impact of Stock Returns on Leverage

The table presents Fama-MacBeth (1973) estimates of parameters and R^2 from Welch's (2004) empirical model of leverage,

$$\frac{D_{t+k}}{D_{t+k} + E_{t+k}} = \alpha_0 + \alpha_1 \cdot \frac{D_t}{D_t + E_t} + \alpha_2 \cdot \frac{D_t}{D_t + E_t \cdot (1 + r_{t,t+k})} + \varepsilon_{t,t+k},$$

where D_t is the firm's debt value, E_t is the firm's equity value, $r_{t,t+k} = (E_{t+k} - E_t)/E_t$ is the equity price appreciation, $\varepsilon_{t,t+k}$ is a random error, and k is the horizon measured in years. The table presents first a reproduction of the results from Panel B of Table 3 in Welch (2004). The second through fourth results use data simulated according to a reduced-form model in which firms rebalance their leverage while facing a fixed cost, a proportional cost, and a fixed plus convex cost of adjustment. The final set of results uses data generated from the dynamic tradeoff model of Fischer, Heinkel, and Zechner (1989). The details of the simulations may be found in Appendix A.

Horizon (k)	α_0	α_1	α_2	R^2
Welch (2004) Results				
1-year	0.03	-0.05	1.02	0.91
3-year	0.07	-0.04	0.94	0.78
5-year	0.09	-0.01	0.87	0.70
10-year	0.14	0.07	0.71	0.56
Simulated Data (Fixed plus Convex Cost)				
1-year	0.04	-0.12	1.02	0.98
3-year	0.09	-0.15	0.90	0.94
5-year	0.13	-0.19	0.83	0.89
10-year	0.21	-0.25	0.68	0.78
Simulated Data (Proportional Cost)				
1-year	0.03	-0.11	1.02	0.99
3-year	0.08	-0.14	0.91	0.95
5-year	0.13	-0.18	0.83	0.92
10-year	0.20	-0.24	0.70	0.82
Simulated Data (Fixed Cost)				
1-year	0.13	-0.36	1.02	0.78
3-year	0.26	-0.39	0.70	0.56
5-year	0.31	-0.37	0.52	0.43
10-year	0.34	-0.24	0.30	0.25
Simulated Data from Fischer, Heinkel, and Zechner (1989) (Fixed Cost)				
1-year	0.05	-0.26	1.11	0.52
3-year	0.16	-0.21	0.66	0.32
5-year	0.19	-0.18	0.49	0.23
10-year	0.22	-0.12	0.33	0.15

(a) Table II: The impact of stock returns on leverage.

Welch-style regression patterns appear in simulated rebalancing models.

Table III
Capital Structure Adjustment Summary Statistics

The sample consists of quarterly COMPUSTAT data from 1984 to 2001 and is restricted to firms with at least four years of contiguous data and no missing values for equity issuances, equity repurchases, long-term debt, short-term debt, or book assets. Financial firms (SICs 6000-6999) and utilities (SICs 4900-4999) are excluded. The table presents summary information on four basic financing spikes (Debt Issue, Debt Retirement, Equity Issue, Equity Repurchase) and two leverage adjustments (Leverage Increase and Leverage Decrease). The basic financing spikes are defined as a net security issuance or repurchase of at least 5% of book assets. The leverage adjustments are defined as a difference in net debt issued and net equity issued that is greater in magnitude than 5% of book assets. The duration measures the time, in quarters, between financing spikes or leverage adjustments of the same type. Right-censored spells are the number of financing spikes or leverage adjustments with a right-censored duration. There are a total of 127,308 firm-quarter observations and the average firm has approximately 36 quarterly observations.

Adjustment Type	Number of Adjustments	Percent of Right-Censored Periods	Median Duration	Adjustments per Firm			
				Mean	Min	Median	Max
No Adjustment	92,159	72.39	-	-	-	-	-
Debt Issue	16,021	12.58	3,114	3	4.19	0	3
Debt Retirement	10,920	8.58	3,087	4	2.80	0	2
Equity Issue	6,867	5.39	3,344	5	1.88	0	1
Equity Repurchase	5,723	4.50	3,390	3	2.81	0	1
Leverage Increase	16,385	12.87	3,122	3	4.23	0	3
Leverage Decrease	15,113	11.87	2,977	4	3.73	0	3

(b) Table III: Capital structure adjustment summary statistics.

Firms are inactive in most quarters but adjust frequently over time.

7.5 Table III and Table IV: Frequency and magnitude of adjustment

Table III read:

- No adjustment occurs in 72.39 percent of firm-quarter observations.
- Debt issues occur in 12.58 percent of quarters; debt retirements in 8.58 percent; equity issues in 5.39 percent; and equity repurchases in 4.50 percent.
- Leverage increases occur in 12.87 percent of quarters, and leverage decreases in 11.87 percent.
- The median duration between debt issues is three quarters; between debt retirements, four quarters; between equity issues, five quarters; and between repurchases, three quarters.
- The typical firm adjusts capital structure roughly once per year, even though most individual quarters are inactive.

Table IV read:

- Median debt issue size is \$7.81 million; median debt retirement size is \$6.62 million.
- Median equity issue size is smaller in dollars at \$3.55 million, while median repurchase size is larger at \$11.20 million.

- Relative to book assets, equity issues are the largest median adjustment at 20 percent of assets; repurchases are much smaller at 3 percent.
- The combination of infrequent but nontrivial actions is consistent with lumpy adjustment costs.

Economic message: Firms are not continuously adjusting each quarter. They exhibit long inactive spells and then make meaningful adjustments, exactly the environment in which duration analysis is useful.

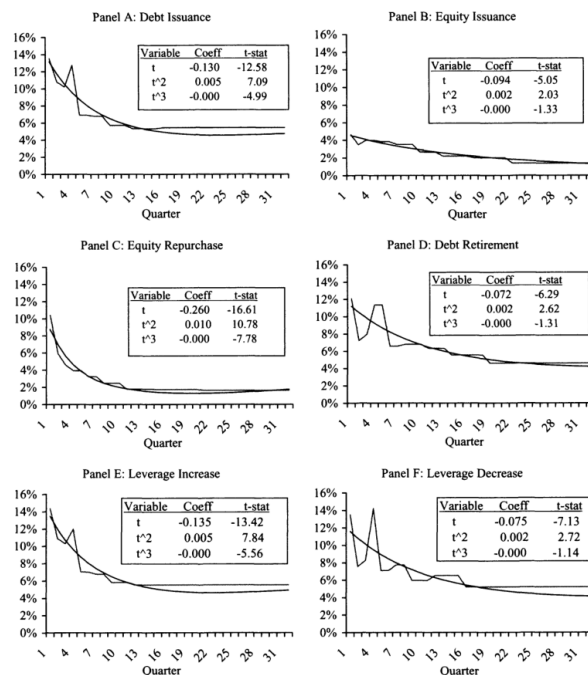
Table IV
Capital Structure Adjustment Magnitudes

The sample consists of quarterly COMPUSTAT data from 1984 to 2001 and is restricted to firms with at least four years of contiguous data and no missing values for equity issuances, equity repurchases, long-term debt, short-term debt, or book assets. Financial firms (SICs 6000–6999) and utilities (SICs 4900–4999) are excluded. The table presents summary information on the magnitude of four basic financing spikes: Debt Issue, Debt Retirement, Equity Issue, and Equity Repurchase, each defined as a net security issuance or repurchase of at least 5% of book assets. All dollar values are in millions and inflation adjusted to 2001 dollars using the all urban CPI. The top one percentile of each variable's distribution is trimmed.

		Median	Mean	Std Dev
Debt issue	Issue size	7.81	54.48	147.66
	Book assets	76.33	504.03	1491.57
	Issue size/Book assets	0.10	0.16	0.17
	Issue size/Market capitalization	0.12	0.23	0.33
Debt retirement	Retirement size	6.62	44.42	120.06
	Book assets	66.12	498.93	1534.71
	Retirement size/book assets	0.09	0.13	0.12
	Retirement size/Market capitalization	0.15	0.40	0.83
Equity issue	Issue size	3.55	19.93	42.42
	Book assets	15.14	154.03	597.65
	Issue size/Book assets	0.20	0.41	0.58
	Issue size/Market capitalization	0.09	0.14	0.17
Equity repurchase	Repurchase size	11.20	55.90	112.46
	Book assets	348.22	1661.63	3142.95
	Repurchase size/Book assets	0.03	0.04	0.03
	Repurchase size/Market capitalization	0.02	0.03	0.04

(a) Table IV: Capital structure adjustment magnitudes.

Adjustment events are economically meaningful and lumpy.



(b) Figure 6: Estimated hazard curves. Actual financing hazards are downward-sloping and clustered.

7.6 Figure 6 and Table V: Duration-model evidence on financing motives

Figure 6 read:

- The estimated hazards for debt issuance, equity issuance, equity repurchase, debt retirement, leverage increases, and leverage decreases are generally downward-sloping.
- Equity repurchase hazards fall especially steeply, consistent with repurchase programs being executed in clusters, partly because of SEC Rule 10b-18 trading-volume restrictions.
- Debt issuance starts at a high hazard and then flattens, indicating frequent clustered debt access.
- Equity issuance has a much lower hazard than debt issuance, consistent with higher equity issuance costs.
- These shapes are most consistent with proportional or fixed-plus-convex adjustment costs rather than purely fixed costs.

Table V read - Panel A:

- Market-to-book raises the probability of leverage decreases much more than leverage increases. This is consistent with valuation effects and equity-related adjustments.
- Cash lowers both leverage-increase and leverage-decrease hazards, consistent with firms using internal funds instead of external financing.
- Profitability lowers the leverage-increase hazard, again consistent with internal funds reducing external financing needs.
- Higher leverage lowers the probability of a leverage increase and raises the probability of a leverage decrease. The hazard impact of leverage is -0.76 percent for leverage increases and +1.60 percent for leverage decreases.
- A positive change in leverage lowers the leverage-increase hazard and raises the leverage-decrease hazard, with symmetric hazard impacts of about 0.57 percent in magnitude.
- LeverDown raises the likelihood of a future leverage increase by 47.03 percent.
- LeverUp raises the likelihood of a future leverage decrease by 65.26 percent.

Table V read - Panel B:

- Debt issuance is less likely when leverage is high and more likely after prior leverage decreases.
- Debt retirement is more likely when leverage is high, when leverage has risen, and after prior leverage increases.
- Equity issuance is strongly related to market-to-book and equity returns, but not strongly to leverage levels.
- Equity repurchases are less likely when leverage is high and more likely when firms have cash and profitability.
- Estimated debt issuance spreads reduce debt issuance likelihood, consistent with issuance costs mattering.
- The equity issuance spread coefficient is positive, which the authors interpret cautiously, possibly due to extrapolating underwriter-spread estimates to smaller firms.

Economic message: The strongest rebalancing channel is debt policy. Firms issue or retire debt in response to leverage deviations and prior leverage movements. Equity issuance contains market-timing behavior but does not imply absence of rebalancing.

Table V
Determinants of Financing Decisions

The sample consists of quarterly COMPUSTAT data from 1984 to 2001 and is restricted to firms with at least four years of contiguous data and no missing values for equity issuances, equity repurchases, long-term debt, short-term debt, or book assets. Financial firms (SICs 6000–6999) and utilities (SICs 4900–4999) are excluded. All variables are normalized by total assets and measured at time $t - 1$, unless otherwise noted, and are defined as follows: *Size* is the ratio of sales for firm i in quarter t to the sum of sales for all firms in quarter t ; *MA/BA* is the ratio of total assets minus book equity plus market equity to total assets; *CapEx* ($t + 1$) is capital expenditures in quarter $t + 1$; *Cash* is cash and short-term marketable securities; *DepAmort* is depreciation and amortization; *Tangibility* is the value of tangible assets; *Profitability* is net operating income; *Volatility* is the absolute value of the change in net income; *Z-score* is the sum of 3.3 times earnings before interest and taxes plus sales plus 1.4 times retained earnings plus 1.2 times working capital, all divided by total assets; *Selling Expense* is selling expenses as a fraction of sales; *Equity Return* is the cumulative four-quarter stock return; Δ *Leverage* is the change in leverage; *Leverage* is the ratio of total debt to the sum of total debt and the market value of equity; *LeverDown* is a binary variable equal to one after a leverage decreasing event occurs during a spell; *LeverUp* is a binary variable equal to one after a leverage increasing event occurs during a spell; *Estimated Spreads* represents the estimated underwriter spread for the issuance that ends each spell, calculated using estimated equations for debt and equity issuance spreads from Altinkilic and Hansen (2000). For right-censored spells, this is replaced by the estimated spread for the issuance or repurchase that ended the previous spell; *Turnover* is the maximum daily turnover during the quarter. Binary variables corresponding to years, quarters, and two-digit SIC codes are included in the estimation but not reported. The hazard impact (HI) is defined as $100 \times (\exp[\beta] - 1)$, where β is the estimated coefficient, and measures the percentage shift in the hazard curve due to a one-unit change in the covariate. Finally, t -statistics are presented in parentheses below the corresponding estimate. Statistical significance at the 1% (5%) level is indicated by two (one) asterisks.

Panel A: Leverage Adjustments				
Coefficient	Leverage Increase		Leverage Decrease	
	Estimate	HI (%)	Estimate	HI (%)
Size	-0.0031** (-2.72)	-0.31	-0.0094** (-5.03)	-0.94
MA/BA	0.0379* (2.5)	3.87	0.1868** (13.06)	20.54
CapEx ($t + 1$)	0.0804** (15.64)	8.37	0.0021 (0.27)	0.21
Cash	-0.0278** (-17.67)	-2.74	-0.0152** (-9.85)	-1.51
DepAmort	-0.06** (-2.94)	-5.82	0.0417 (1.92)	4.26
Tangibility	-0.0034** (-3.27)	-0.34	-0.0116** (-9.27)	-1.16
Profitability	-0.0245** (-5.29)	-2.42	-0.0004 (-0.1)	-0.04
Volatility	0.0072 (1.42)	0.72	0.0221** (4.53)	2.24
Z-score	0.0000 (0.24)	0.00	-0.0008** (-6.6)	-0.08
Selling expense	0.0004 (0.62)	0.04	0.0013* (2.01)	0.13
Equity return	0.0004 (1.37)	0.04	0.0023** (8.05)	0.23

(continued)

Table V—Continued

Panel A: Leverage Adjustments								
Coefficient	Leverage Increase			Leverage Decrease				
	Estimate	HI (%)		Estimate	HI (%)		Estimate	HI (%)
Δ Leverage	-0.0057** (-2.94)	-0.57		0.0057** (2.82)			0.57	
Leverage	-0.0076** (-8.83)	-0.76		0.0159** (18.02)			1.60	
LeverDown	0.3855** (12.03)	47.03						
LeverUp				0.5023** (12.77)			65.26	

Panel B: Basic Financing Spikes								
Coefficient	Debt Issuance		Equity Issuance		Debt Retirement		Equity Repurchase	
	Estimate	HI (%)	Estimate	HI (%)	Estimate	HI (%)	Estimate	HI (%)
Size	-0.0079** (-5.64)	-0.79	-0.0358** (-4.62)	-3.52	-0.0073** (-3.55)	-0.73	0.0063** (5.21)	0.63
MA/BA	0.0206 (1.29)	2.08	0.1934** (10.28)	21.33	0.1098** (5.03)	11.61	0.0431 (1.64)	4.40
CapEx ($t + 1$)	0.0869** (16.83)	9.08	0.0499** (5.08)	5.12	-0.0287** (-2.76)	-2.83	-0.0374* (-2.53)	-3.67
Cash	-0.0359** (-19.93)	-3.52	-0.0161** (-7.1)	-1.59	-0.0204** (-8.97)	-2.02	0.0084** (4.43)	0.84
DepAmort	-0.06** (-2.9)	-5.82	0.0717* (2.09)	7.43	0.0241 (0.88)	2.44	-0.0138 (-0.32)	-1.37
Tangibility	-0.004** (-3.75)	-0.40	-0.0054* (-2.55)	-0.54	-0.0134** (-8.77)	-1.33	0.0032 (1.55)	0.32
Profitability	-0.0313** (-6.82)	-3.08	-0.0265** (-3.7)	-2.62	0.0129* (2.19)	1.30	0.0305* (2.4)	3.10
Volatility	0.0029 (0.56)	0.29	-0.0063 (-0.72)	-0.62	0.0357** (6.13)	3.64	0.0065 (0.51)	0.65
Z-score	-0.0003* (-2.38)	-0.03	-0.0012** (-6.91)	-0.12	-0.0004* (-2.19)	-0.04	0.0015** (3.12)	0.15
Sell Exp	0.0007 (1.16)	0.07	0.0004 (0.48)	0.04	0.0002 (0.21)	0.02	-0.004* (-2.28)	-0.40
Equity return	0.0009** (3.48)	0.09	0.0051** (11.28)	0.52	0.0015** (4.21)	0.15	-0.0014** (-2.59)	-0.14
Δ Leverage	-0.0044* (-2.26)	-0.44	-0.0071 (-1.65)	-0.71	0.0063** (2.79)	0.63	-0.0024 (-0.49)	-0.24
Leverage	-0.0081** (-9.07)	-0.80	0.0020 (1.06)	0.20	0.0181** (17.01)	1.83	-0.018** (-8.95)	-1.78
LeverDown	0.3675** (11.41)	44.41					-0.0177 (-0.34)	-1.76
LeverUp			0.2525** (3.68)	28.73	0.5439** (12.24)	72.28		
Estimated spread	-0.0018** (-2.96)	-0.18	0.0027** (3.23)	0.27				
Turnover							0.0004 (1.43)	0.04

(a) Table V: Determinants of financing decisions, first page.

Panel A shows the leverage-adjustment hazard estimates.

(b) Table V continued: leverage adjustments and basic financing spikes.

Debt policy provides the clearest evidence of active rebalancing.

8 Main empirical conclusions

8.1 Adjustment costs matter

- Firms are inactive in almost three-quarters of firm-quarter observations.
- Financing events cluster in time once firms access capital markets.
- Hazard curves are generally downward-sloping, suggesting clustered adjustments and cost structures closer to proportional or fixed-plus-convex costs.
- Estimated debt issuance costs reduce the debt issuance hazard.
- These facts are inconsistent with a frictionless partial-adjustment model but consistent with costly adjustment.

8.2 Firms actively rebalance

- Firms with higher leverage are more likely to reduce leverage and less likely to increase it.

- Firms whose leverage has increased are more likely to reduce leverage.
- Firms whose leverage has decreased are more likely to increase leverage.
- Past leverage decreases make future leverage increases more likely; past leverage increases make future leverage decreases more likely.
- The magnitude of these effects is economically large: prior leverage decreases raise leverage-increase hazards by 47 percent, and prior leverage increases raise leverage-decrease hazards by 65 percent.

8.3 Rebalancing is mainly through debt policy

- Debt issuances and retirements respond strongly to leverage levels, leverage changes, and prior leverage adjustments.
- Equity repurchases also play some role in rebalancing.
- Equity issuances are more strongly tied to market-to-book and recent stock returns than to leverage deviations.
- Therefore, equity market timing can coexist with active rebalancing: firms may time equity issuance and later offset leverage effects with debt.

8.4 Pecking-order and tradeoff elements coexist

- Cash and profitability reduce the likelihood of external financing, consistent with pecking-order logic.
- Capital expenditures increase the likelihood of debt and equity issuance, consistent with investment-financing needs.
- High leverage raises the probability of debt retirement, consistent with bankruptcy-cost and target-range considerations.
- The evidence therefore supports a dynamic target-range view with adjustment costs, while also leaving room for adverse-selection and internal-funds motives.

9 Relation to prior capital-structure theories

9.1 Against a strong market-timing interpretation

The paper does not claim that firms never time equity markets. Equity issuances are clearly related to market-to-book and stock returns. The claim is instead that market timing does not imply permanent indifference to leverage. Firms that issue equity often subsequently issue debt, and the EFWA effect weakens when adjustment costs are lower.

9.2 Against a strong inertia interpretation

Welch's inertia argument rests on the persistence of equity-return shocks in leverage. Leary and Roberts show that similar regression patterns arise from simulated dynamic tradeoff models with adjustment costs. Thus, Welch-style regressions are not powerful enough to distinguish inertia from costly rebalancing.

9.3 Connection to dynamic tradeoff theory

The evidence is consistent with a dynamic tradeoff model in which firms tolerate leverage inside a target range and adjust when leverage approaches boundaries. The target is not necessarily a point; it is better understood as a band.

9.4 Connection to modified pecking order

The evidence also fits a modified pecking-order view in which internal funds are preferred, but leverage constraints and bankruptcy costs matter. Firms do not always issue debt simply because it is cheaper than equity; they are more careful when leverage is high.

10 Short summary

- Leary and Roberts ask whether firms rebalance capital structures after accounting for costly adjustment.
- Adjustment costs imply no-adjustment regions, lumpy financing, clustered adjustments, and persistent leverage shocks even with active rebalancing.
- The authors show that Baker-Wurgler market-timing evidence and Welch inertia evidence can be reinterpreted through the lens of adjustment costs.
- Quarterly data reveal that firms make no capital-structure adjustment in 72 percent of quarters, but still adjust about once per year.
- A duration model estimates the hazard of debt issues, debt retirements, equity issues, repurchases, leverage increases, and leverage decreases.
- The hazard shapes and cost proxies support the presence of adjustment costs.
- Rebalancing variables strongly predict financing decisions: high leverage predicts leverage decreases; low or falling leverage predicts leverage increases.
- Past leverage decreases raise the likelihood of future leverage increases; past leverage increases raise the likelihood of future leverage decreases.
- Debt policy is the main rebalancing tool; equity issuances are more related to market-to-book and equity returns.
- The paper concludes that persistent leverage effects are more likely due to optimizing behavior with adjustment costs than indifference toward capital structure.

11 Conclusion

11.1 Core conclusion

Firms do rebalance their capital structures, but they do not do so continuously. They behave as if they have an optimal leverage range and face meaningful costs of adjusting capital structure. Because of these costs, firms allow leverage shocks to persist until the benefits of adjustment justify a financing action.

11.2 Economic intuition

A firm that is slightly underlevered or overlevered may rationally wait. The gain from moving leverage closer to target can be smaller than the fixed, proportional, or convex cost of issuing or retiring securities. Once the deviation becomes large enough, the firm adjusts, often in clusters. Therefore, persistent shocks are not evidence that managers ignore capital structure; they can be evidence that managers optimize under frictions.

11.3 Incremental contribution revisited

The paper's key contribution is to change the empirical object from slow leverage movements to discrete financing decisions. By studying the duration between adjustment events and the determinants of those events, the paper shows that firms' actual financing choices are strongly linked to leverage deviations and prior leverage movements.

11.4 Implications

- **For empirical capital-structure work.** Continuous partial-adjustment models can underestimate the speed and importance of rebalancing when actual adjustment is lumpy.
- **For interpreting market timing.** Equity market timing can exist, but its leverage effects need not be permanent.
- **For interpreting stock-return effects.** Equity returns mechanically move leverage, but firms respond through debt issuance and retirement over subsequent quarters.
- **For theory.** A realistic model of capital structure should combine target ranges, adjustment costs, internal funds, investment needs, and financing frictions.

11.5 Limitations

- The duration model is reduced form; it does not estimate a fully structural dynamic capital-structure model.
- Adjustment-cost proxies are imperfect, especially for equity issuance costs and debt retirement costs.
- The paper cannot fully separate a dynamic tradeoff model from a modified pecking-order model, because both can predict delayed responses to leverage deviations.
- The event classifications use thresholds, although robustness checks with alternative thresholds give similar results.

11.6 Takeaway

The paper's main lesson is that firms appear to care about leverage, but they care about adjustment costs too. Capital structures are not continuously reset to a point target. They evolve inside a range, with active but intermittent rebalancing.